

Olympiad Test Paper — Science (Class 7)

Chapter 9: Soil

Subject	Science (NCERT Class 7)
Chapter	9 — Soil
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. For percolation problems use the rule $\text{percolation rate} = \text{volume of water} \div \text{time taken (mL/min)}$. No calculator is needed — all numbers are designed to work by hand. Most questions need two to four reasoning steps; watch for traps. Do not write on this paper — mark answers on your answer sheet.

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- The layered top-to-bottom slice of soil, showing all its horizons, is called the:
(A) soil texture (B) percolation rate
(C) soil profile (D) humus layer
 - Which horizon of a soil profile is the dark, humus-rich topsoil where most plant roots grow?
(A) the C horizon (B) the R horizon
(C) the B horizon (D) the A horizon
 - Arrange the soil horizons in order from the surface downward.
(A) bedrock, C, B, A, O (B) A, O, B, C, bedrock
(C) O, A, B, C, bedrock (D) O, B, A, C, bedrock
 - Soil is formed mainly by the:
(A) sudden cooling of lava in one night (B) freezing of rainwater in winter
(C) slow weathering of rocks over a very long time
(D) rapid evaporation of river water
 - The dark, crumbly material formed from decayed plant and animal remains in the top layer of soil is called:
(A) clay (B) silt
(C) bedrock (D) humus

- 6.** A gardener digs a pit and finds, just under the topsoil, a lighter, harder layer where minerals washed down from above have collected. This layer is the:
- (A) A horizon (B) C horizon
(C) O horizon (D) B horizon
- 7.** The solid, unweathered rock at the very bottom of a soil profile is the:
- (A) subsoil (B horizon) (B) bedrock (R horizon)
(C) topsoil (A horizon) (D) organic layer (O horizon)
- 8.** Which statement about how soil forms is correct?
- (A) Soil forms in a few days after heavy rain.
(B) Soil is made entirely of living organisms. (C) Soil never contains any rock material.
(D) Building a centimetre of new topsoil can take hundreds of years.
- 9.** A desert soil often shows almost no O horizon. The most likely reason is:
- (A) too much rainfall washes the O horizon away
(B) wind packs the particles too tightly (C) deserts have no bedrock underneath
(D) there is very little plant litter to form humus
- 10.** Which of these correctly pairs a soil type with its particle size?
- (A) clayey — largest particles (B) sandy — smallest particles
(C) sandy — largest particles (D) loamy — only one particle size
- 11.** Water rushes straight through sandy soil mainly because:
- (A) the large gaps between big particles let water drain fast
(B) sand particles dissolve in water (C) sand repels water
(D) sand has no air spaces at all
- 12.** Which soil holds the most water and drains the slowest?
- (A) sandy soil (B) clayey soil
(C) loamy soil (D) silt alone
- 13.** Loamy soil is the best for growing most crops because it:
- (A) is made only of large sand grains
(B) holds enough water for roots yet lets the extra drain away
(C) never contains any humus (D) is the same as pure clay
- 14.** Two pots are filled to the same level — one with sandy soil, one with clayey soil — and equal water is poured on each. After ten minutes the:
- (A) sandy pot has drained more water out of the bottom
(B) clayey pot has drained more water out of the bottom
(C) both drain exactly the same amount (D) neither lets any water through
- 15.** Black cotton soil of the Deccan plateau, which becomes very sticky when wet and cracks when dry, is mainly:

- (A) sandy soil
- (B) pure silt
- (C) loose gravel
- (D) clayey soil

16. Which property best explains why sandy soil dries out quickly after rain?

- (A) high percolation rate and low water-holding capacity
- (B) high water-holding capacity
- (C) very small air gaps
- (D) very high humus content

17. Soil is described as "teeming with life." Which best supports this statement?

- (A) Soil is only broken rock with nothing living in it.
- (B) A single gram of soil can contain millions of micro-organisms.
- (C) Only earthworms live in soil and nothing else.
- (D) Living things avoid soil completely.

18. Roughly, the four main ingredients of soil are:

- (A) only sand and water
- (B) mineral particles, air, water, and humus
- (C) plastic, glass, metal, and stone
- (D) only humus and air

19. When a soil is completely water-logged after heavy rain, plant roots can struggle because the soil now has very little:

- (A) air
- (B) water
- (C) clay
- (D) weight

20. Which agent is **not** normally responsible for weathering rock into soil over time?

- (A) a single bolt of lightning
- (B) flowing and freezing water
- (C) growing plant roots
- (D) wind

21. A student heats a sample of moist soil in a dry test tube and sees droplets form on the cooler upper walls. This shows that soil contains:

- (A) plastic
- (B) air
- (C) moisture (water)
- (D) salt only

22. To show that soil contains **air**, the best simple demonstration is to:

- (A) drop a lump of dry soil into water and watch bubbles rise
- (B) heat the soil until droplets appear on the glass
- (C) rub the soil between your fingers
- (D) weigh the soil on a balance

23. Percolation rate is best defined as the:

- (A) weight of soil per unit volume
- (B) amount of humus in the soil
- (C) number of organisms in the soil
- (D) speed at which water seeps down through soil

24. In an experiment, 200 mL of water takes 20 minutes to percolate through a soil sample. The percolation rate is:

- (A) 0.1 mL/min (B) 10 mL/min
(C) 4000 mL/min (D) 220 mL/min

25. 150 mL of water percolates through a soil column in 5 minutes. The percolation rate is:

- (A) 30 mL/min (B) 750 mL/min
(C) 3 mL/min (D) 145 mL/min

26. Soil X lets 300 mL through in 10 minutes; soil Y lets 300 mL through in 30 minutes. Which soil is more likely sandy?

- (A) soil Y, because it is slower (B) both are equally sandy
(C) neither — sand never lets water through
(D) soil X, because it has the higher percolation rate

27. A loamy sample drains 240 mL of water in 8 minutes. Its percolation rate is:

- (A) 248 mL/min (B) 30 mL/min
(C) 0.03 mL/min (D) 1920 mL/min

28. A sandy soil has a percolation rate of 25 mL/min. How long will 200 mL of water take to percolate through it?

- (A) 8 minutes (B) 5000 minutes
(C) 0.125 minutes (D) 175 minutes

29. Soil P percolates 200 mL in 8 minutes; soil Q percolates 360 mL in 12 minutes. Which has the **higher** percolation rate?

- (A) soil Q (B) soil P
(C) both equal (D) cannot be compared

30. Two soils drain equal volumes of water. Sandy soil takes 4 minutes; clayey soil takes 40 minutes. The ratio of their percolation rates (sandy : clayey) is:

- (A) 1 : 10 (B) 4 : 40
(C) 10 : 1 (D) 1 : 1

31. Water-holding capacity is **highest** in:

- (A) clayey soil (B) sandy soil
(C) gravel (D) coarse sand

32. Which soil type usually has the **best aeration** (most air for roots)?

- (A) water-logged soil (B) clayey soil
(C) sandy soil (D) compacted clay

33. Paddy (rice) is grown in fields that are kept flooded with water. The most suitable soil for paddy is therefore:

- (A) sandy soil, because it drains fast (B) gravel, because it has big gaps
(C) clayey soil, because it holds water (D) dry desert soil

34. Cotton grows best in:

- (A) water-logged clay that never drains (B) solid bedrock
(C) pure beach sand with no nutrients (D) loamy or sandy-loam soil rich in humus

35. A farmer's field stays water-logged for days after every rain, and crops keep rotting. To improve drainage he should mix in some:

- (A) more fine clay (B) plastic sheeting
(C) sand (D) extra water

36. Which crop is best matched to a soil that holds water well and is poorly aerated?

- (A) groundnut in dry sand (B) paddy (rice)
(C) cactus (D) watermelon on a sand dune

37. A cactus naturally thrives in:

- (A) water-logged clay (B) flooded paddy fields
(C) thick humus bog (D) well-drained sandy soil

38. Which sequence of soils is in **increasing** order of percolation rate (slowest to fastest)?

- (A) clayey, loamy, sandy (B) sandy, loamy, clayey
(C) loamy, sandy, clayey (D) sandy, clayey, loamy

39. The wearing-away and removal of fertile topsoil by wind or water is called:

- (A) percolation (B) weathering
(C) crop rotation (D) soil erosion

40. Which human activity speeds up soil erosion the most?

- (A) planting trees along field edges (B) cutting down forests on hill slopes
(C) building check-dams in gullies (D) ploughing along the contour

41. Cutting flat, step-like platforms into a hillside to slow water and stop soil washing down is called:

- (A) contour ploughing (B) crop rotation
(C) overgrazing (D) terracing

42. Ploughing **along** the elevation lines of a slope, so each furrow traps water instead of channelling it downhill, is called:

- (A) contour ploughing (B) terracing
(C) deforestation (D) mulching

43. Why does keeping a field covered with plants or mulch reduce soil erosion?

- (A) The cover makes the soil heavier so it cannot move.
(B) Mulch turns soil into solid rock.
(C) The cover stops all rain from ever reaching the ground.
(D) Plant roots and mulch break the force of rain and hold the soil in place.

44. Rows of trees planted to slow down wind and protect a field from wind erosion are called:

- (A) check-dams
- (B) terraces
- (C) shelter belts
- (D) furrows

45. Crop rotation helps keep soil fertile mainly because:

- (A) it removes all the soil's water
- (B) it makes the soil purely sandy
- (C) growing legumes restores nitrogen taken out by cereal crops
- (D) it stops the soil from ever being ploughed

46. In a cereal–legume–cereal rotation, the legume (such as gram or peas) helps because bacteria in its roots:

- (A) fix nitrogen from the air into the soil
- (B) eat up the soil's nitrogen
- (C) turn the soil into clay
- (D) increase the percolation rate to zero

47. Which of these is a form of **soil pollution**?

- (A) earthworms moving through the soil
- (B) plastic waste and harmful chemicals dumped on the soil
- (C) rain percolating through the soil
- (D) humus forming from fallen leaves

48. Deforesting a steep hillside is especially harmful because, with the tree roots gone:

- (A) the soil becomes more fertile
- (B) percolation completely stops
- (C) heavy rain washes the loose topsoil away easily
- (D) the soil holds more water than before

49. A field of bare, dry soil on a windy plain is most at risk from:

- (A) water-logging
- (B) wind erosion
- (C) too much humus
- (D) crop rotation

50. Why is losing topsoil to erosion such a serious problem for farming?

- (A) Topsoil regrows fully within a week.
- (B) Topsoil has no nutrients anyway.
- (C) Topsoil is the humus-rich layer where crops feed, and it takes centuries to rebuild.
- (D) Erosion only removes harmless bedrock.

51. A farmer wants a soil that holds enough water for vegetables but still drains after rain and is easy to till. He should choose:

- (A) pure sand
- (B) loamy soil
- (C) pure clay
- (D) solid bedrock

52. Equal volumes of water are poured on three samples. Sandy drains in 2 min, loamy in 6 min, clayey in 18 min. Which has the **lowest** percolation rate?

- (A) loamy
- (B) clayey
- (C) sandy
- (D) all equal

- 53.** A soil column drains 300 mL in 6 minutes; a second column of the same soil drains 600 mL. Assuming the same rate, the second takes:
- (A) 12 minutes (B) 3 minutes
(C) 1 minute (D) 6 minutes
- 54.** Which statement about loamy soil is **false**?
- (A) It is a mixture of sand, silt, clay and humus.
(B) It has the slowest percolation rate of all soils.
(C) It is well suited to most crops. (D) It holds water moderately well.
- 55.** A gram of garden soil supports more living organisms than a gram of beach sand mainly because garden soil:
- (A) is heavier (B) has larger particles
(C) contains no air
(D) is rich in humus and holds moisture for organisms
- 56.** Soil A drains 100 mL in 4 min; soil B drains 100 mL in 4 min as well, but soil B's sample is twice as deep. The slower-percolating arrangement, per unit depth, is:
- (A) soil B (B) soil A
(C) both identical per unit depth (D) neither lets water through
- 57.** Why does over-watering a clayey-soil pot often harm the plant?
- (A) Clay drains so fast the roots dry out. (B) Clay has too much air already.
(C) Clay holds the extra water and starves the roots of air.
(D) Clay turns the water into humus.
- 58.** A hill farmer plants crops in straight up-and-down rows on a steep slope and clears all the trees. After the monsoon his field has deep channels and lost soil. The single best change he could make is to:
- (A) plough up-and-down even more (B) cut down the remaining vegetation
(C) pour extra water on the slope
(D) terrace the slope and plough along the contour
- 59.** Which best describes the role of percolation for groundwater?
- (A) Percolation removes all water from the ground.
(B) Water that percolates down helps refill underground water stores.
(C) Percolation only matters in deserts. (D) Percolation has no effect on groundwater.
- 60.** A soil sample lets 480 mL of water through in 12 minutes. At the same rate, how much water would percolate in 5 minutes?
- (A) 40 mL (B) 2400 mL
(C) 200 mL (D) 96 mL

Solutions — Science (Class 7), Chapter 9: Soil

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	C	31	A	41	D	51	B
2	D	12	B	22	A	32	C	42	A	52	B
3	C	13	B	23	D	33	C	43	D	53	A
4	C	14	A	24	B	34	D	44	C	54	B
5	D	15	D	25	A	35	C	45	C	55	D
6	D	16	A	26	D	36	B	46	A	56	A
7	B	17	B	27	B	37	D	47	B	57	C
8	D	18	B	28	A	38	A	48	C	58	D
9	D	19	A	29	A	39	D	49	B	59	B
10	C	20	A	30	C	40	B	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (C)** The complete vertical top-to-bottom section through soil, showing all its horizons, is the soil profile. ("Texture" is about particle size; "percolation rate" is a drainage measure.)
- (D)** The A horizon is the topsoil — dark, rich in humus and minerals, and the layer where most roots grow. The O horizon is the surface litter, B is the lighter subsoil, C is broken parent rock.
- (C)** From the surface downward the horizons run O (organic litter), A (topsoil), B (subsoil), C (weathered rock), then bedrock — so O, A, B, C, bedrock.
- (C)** Soil forms by the slow weathering of rocks — water, wind, ice, and living things break rock into fine particles over a very long time. It is a gradual process, not a sudden one.
- (D)** Humus is the dark, crumbly material formed from decayed plant and animal remains; it makes the topsoil fertile. Clay and silt are mineral particles; bedrock is solid rock.
- (D)** Just below the topsoil (A horizon) lies the B horizon (subsoil) — lighter and harder, where clay and minerals washed down from above accumulate.

- 7. (B)** The bottom of a soil profile is the unweathered solid rock, the bedrock or R horizon. Topsoil (A horizon), subsoil (B horizon) and the organic layer (O horizon) all lie above it.
- 8. (D)** Soil forms extremely slowly — building even one centimetre of new topsoil can take hundreds of years. Soil is not made entirely of living things, it does contain rock-derived mineral particles, and it certainly does not form in a few days.
- 9. (D)** The O horizon is built from plant litter turning into humus. Deserts have very few plants, so there is little litter — hence almost no O horizon. (It is not washed away by rain — deserts get little rain.)
- 10. (C)** Sandy soil has the largest particles, clayey soil the smallest, and loamy soil is a mixture of several particle sizes. So the correct pairing is "sandy — largest particles."
- 11. (A)** Big sand particles leave large gaps (pores) between them, so water drains straight through quickly. Sand does not dissolve, does not repel water, and it does have air spaces.
- 12. (B)** Clayey soil has the tiniest, most tightly packed particles, so it drains slowest and holds the most water. Sandy drains fastest; loamy is in between.
- 13. (B)** Loam is a balanced mix of sand, silt, clay and humus, so it holds enough water for roots yet still lets surplus water drain — ideal for most crops. It is not pure clay and does contain humus.
- 14. (A)** Sandy soil has a high percolation rate, so far more water drains out of the bottom of the sandy pot than the clayey one in the same ten minutes. (The clayey pot holds the water and lets little through.)
- 15. (D)** Black cotton soil (Deccan plateau) is sticky when wet, cracks when dry, and holds water — these are the signs of clayey soil.
- 16. (A)** Sandy soil drains fast (high percolation rate) and holds little water (low water-holding capacity), so it loses its moisture quickly and dries out soon after rain.
- 17. (B)** Soil is "alive" because a single gram can hold millions of micro-organisms (plus worms, insects, fungi). The other options deny that soil contains life.
- 18. (B)** The four main ingredients of soil are mineral particles, air, water, and humus (organic matter). Plastic/glass/metal are pollutants, not natural soil components.
- 19. (A)** When soil is water-logged, water fills the pore spaces and pushes the air out, so the roots get very little air (oxygen) — which is why they struggle.
- 20. (A)** Weathering is the slow break-up of rock by water (flowing and freezing), wind, and growing roots over long times. A single bolt of lightning is a one-off event, not a normal soil-forming weathering agent.

- 21. (C)** Heating moist soil drives off water as vapour, which condenses into droplets on the cooler glass — proof that the soil contained moisture (water).
- 22. (A)** Dropping a lump of dry soil into water makes bubbles rise as water fills the pore spaces and pushes the trapped air out — a simple proof that soil contains air. (Heating shows water; weighing and rubbing show neither directly.)
- 23. (D)** Percolation rate is the speed at which water seeps down through soil (volume of water $\div$ time). It is not a weight, a humus measure, or an organism count.
- 24. (B)** Percolation rate = volume $\div$ time = 200 mL $\div$ 20 min = 10 mL/min. (Multiplying instead of dividing gives the trap 4000.)
- 25. (A)** Rate = 150 mL $\div$ 5 min = 30 mL/min. (750 = 150 $\times$ 5, the multiply trap; 145 = 150 – 5, the subtract trap.)
- 26. (D)** Same volume (300 mL): soil X takes 10 min, soil Y takes 30 min. X is faster, so X has the higher percolation rate — the mark of sandy soil. (Slower = clayey.)
- 27. (B)** Rate = 240 mL $\div$ 8 min = 30 mL/min. (248 = 240 + 8 is the add trap.)
- 28. (A)** Time = volume $\div$ rate = 200 mL $\div$ 25 mL/min = 8 minutes. (Multiplying 200 $\times$ 25 gives the trap 5000.)
- 29. (A)** Rate of P = 200 $\div$ 8 = 25 mL/min; rate of Q = 360 $\div$ 12 = 30 mL/min. Q (30) > P (25), so soil Q has the higher rate.
- 30. (C)** For equal volumes, rate is inversely proportional to time. Sandy : clayey rate = (1/4) : (1/40) = 10 : 1. (Writing 4 : 40 = 1 : 10 reverses it — the classic trap.)
- 31. (A)** Clayey soil, with its tiny tightly-packed particles, holds the most water — the highest water-holding capacity. Sand, gravel and coarse sand all drain freely.
- 32. (C)** Sandy soil has large air-filled gaps between its big particles, so it is the best aerated. Clayey, compacted and water-logged soils all squeeze out the air.
- 33. (C)** Paddy fields are kept flooded, so the soil must retain water — clayey soil, which holds water well, suits paddy best. Sandy soil and gravel would drain the water away.
- 34. (D)** Cotton grows best in well-drained but humus-rich loamy or sandy-loam soil — enough water plus good aeration. Water-logged clay, bedrock and nutrient-free sand are all unsuitable.
- 35. (C)** Mixing in sand opens up larger pore spaces, raising the percolation rate so the water-logged clay drains better. Adding more clay or water makes it worse; plastic sheeting blocks drainage.

- 36. (B)** A soil that holds water well but is poorly aerated is clayey soil, and the crop matched to flooded, water-holding soil is paddy (rice). Cactus, groundnut and watermelon all prefer dry, sandy conditions.
- 37. (D)** A cactus is adapted to dry conditions and thrives in well-drained sandy soil. Water-logged clay, flooded paddy fields and humus bogs would rot its roots.
- 38. (A)** Percolation rate increases (slowest to fastest) as particle size grows: clayey (slowest) < loamy (medium) < sandy (fastest). So clayey, loamy, sandy.
- 39. (D)** The wearing-away and removal of fertile topsoil by wind or water is soil erosion. Percolation is water seeping down; weathering forms soil; crop rotation maintains fertility.
- 40. (B)** Cutting down forests on hill slopes removes the roots that bind the soil, so rain washes the topsoil away — the biggest erosion-accelerator listed. Planting trees, check-dams and contour ploughing all reduce erosion.
- 41. (D)** Cutting flat, step-like platforms into a hillside is terracing; it slows water and stops soil washing down. Contour ploughing follows the slope's lines; crop rotation and overgrazing are unrelated.
- 42. (A)** Ploughing along the slope's elevation (contour) lines so each furrow traps water is contour ploughing. Terracing cuts steps; deforestation and mulching are different practices.
- 43. (D)** Plant roots bind the soil and the leafy cover/mulch breaks the force of falling raindrops, so the soil is held in place and washes away less. The cover does not turn soil to rock or block all rain.
- 44. (C)** Rows of trees planted to slow the wind and shield a field from wind erosion are shelter belts. Check-dams trap runoff in gullies; terraces and furrows are ground-shaping methods.
- 45. (C)** Crop rotation keeps soil fertile because growing legumes restores the nitrogen that cereal crops remove, so fertility is renewed without heavy fertiliser use.
- 46. (A)** Legumes host Rhizobium bacteria in their root nodules; these bacteria fix nitrogen from the air into the soil as usable nitrates — the opposite of eating it up.
- 47. (B)** Dumping plastic waste and harmful chemicals on the soil is soil pollution. Earthworms, percolating rain and humus from fallen leaves are all natural and beneficial.
- 48. (C)** Once the tree roots that held the soil are gone, heavy rain on the bare hillside easily washes the loose topsoil away. Deforestation lowers fertility and water retention, not raises them.

- 49. (B)** Bare, dry soil on a windy plain has no cover and loose fine particles, so it is most at risk from wind erosion. (Dry soil is not at risk of water-logging.)
- 50. (C)** The topsoil is the humus-rich layer where crops get their nutrients, and because soil forms so slowly it takes centuries to rebuild — so losing it is a serious, near-permanent loss.
- 51. (B)** A soil that holds enough water for vegetables yet still drains and is easy to till is loamy soil — the balanced mix. Pure sand drains too fast, pure clay water-logs, and bedrock is not soil.
- 52. (B)** For equal volumes, the soil that takes the longest time has the lowest rate. Clayey takes 18 min (longest), so clayey has the lowest percolation rate. Sandy (2 min) is fastest.
- 53. (A)** Rate = $300 \text{ mL} \div 6 \text{ min} = 50 \text{ mL/min}$. Time for 600 mL = $600 \div 50 = 12 \text{ minutes}$. (Doubling the volume doubles the time, from 6 to 12 min.)
- 54. (B)** Loamy soil does NOT have the slowest percolation rate — clayey soil does; loam drains moderately. The other three statements (a mix of sand/silt/clay/humus, suited to most crops, holds water moderately) are all true, so the false one is option (B).
- 55. (D)** Garden soil is rich in humus and holds moisture, providing food and water for organisms, so it supports far more life than dry, nutrient-poor beach sand. Weight and particle size are not the reason.
- 56. (A)** Both drain 100 mL in 4 min, so both have the same overall rate — but soil B's column is twice as deep, so per unit depth the water moves more slowly through it. Hence soil B is the slower-percolating arrangement per unit depth.
- 57. (C)** Clayey soil holds the extra water in its tiny pores, pushing out the air, so the roots are starved of oxygen and the plant suffers. (Clay drains slowly, not fast, and does not turn water into humus.)
- 58. (D)** The damage comes from up-and-down ploughing on a bare, deforested slope. The best fix is to terrace the slope and plough along the contour, so water is trapped instead of channelling downhill. Ploughing up-and-down more or clearing more vegetation makes it worse.
- 59. (B)** Water that percolates downward through the soil seeps into the ground and helps refill underground water stores (aquifers). It does not remove all groundwater, and it matters everywhere, not just in deserts.
- 60. (C)** Rate = $480 \text{ mL} \div 12 \text{ min} = 40 \text{ mL/min}$. In 5 min the volume = $40 \times 5 = 200 \text{ mL}$. ($2400 = 480 \times 5$ is the multiply trap; $96 = 480 \div 5$ ignores the rate.)
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Solutions complete: 60 / 60.